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# TAN016 - Tracker One USB charging current advisory

Advisory notice to all Tracker One customers to review the USB cable, USB power source, and USB charging/input current settings used with Tracker One devices.

This advisory applies to Tracker One devices that are powered or charged using the external USB-C connector, including development, test, and production deployments. Tracker One setup documentation describes use of an included USB-C to USB-A cable, or a customer-supplied USB-C to USB-C cable when the computer includes a USB-C port.

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## Issue summary

Particle is issuing this advisory to all Tracker One customers because Tracker One devices may be used with USB-A to USB-C cables, USB chargers, vehicle adapters, extension cables, or other USB power accessories selected by the customer or replaced in the field.

The USB cable and power source used with Tracker One must be appropriate for the charging/input current configured on the device. The Tracker One development package includes a USB-A to USB-C cable for setup and development. In production, customers are responsible for selecting, qualifying, installing, and maintaining USB cables and power sources suitable for their intended use, environment, and configured charging current. They may purchase USB cables from Particle or they may source their own. It is critical however that a customer deployment utilizes an appropriate USB cable including power capabilities that match the Tracker One input current capabilities.

If an under-rated, damaged, worn, long, high-resistance, or otherwise unsuitable USB cable or connector is used with Tracker One, the higher charging/input current that occurs on the cable may cause excessive heating at the USB connector or cable. Heating is dependent on the current, cable resistance, connector condition, and installation environment. The power dissipated as heat in a resistive cable or connector path increases approximately with the square of current.

If USB connector heating, discoloration, deformation, odor, intermittent charging, or charge faults are observed, discontinue use of that cable and power source until the installation has been inspected and corrected.

## Products affected

This advisory applies to all Tracker One customers and all Tracker One devices that use the external USB-C connector for power or battery charging, including but not limited to:

| SKU       | Description                                           | Region | Lifecycle |
|-----------|-------------------------------------------------------|--------|-----------|
| ONE404MEA | Tracker One LTE M1 (NorAm, EtherSIM), [x1]            | NORAM  | GA        |
| ONE404MTY | Tracker One LTE M1 (NorAm, EtherSIM), Bulk [x40]      | NORAM  | GA        |
| ONE523MEA | Tracker One LTE CAT1/3G/2G (Europe), [x1]             | EMEAA  | GA        |
| ONE523MTY | Tracker One CAT1/3G/2G (Europe), Bulk [x40]           | EMEAA  | GA        |
| ONE524MEA | Tracker One LTE CAT1/3G/2G (Europe, EtherSIM), [x1]   | EMEAA  | GA        |
| ONE524MTY | Tracker One CAT1/3G/2G (Europe, EtherSIM), Bulk [x40] | EMEAA  | GA        |

Tracker One devices powered only through the M8/VIN power input and not charged or powered by USB are not exposed to this specific USB connector heating condition. However, customers are still responsible for qualifying the external power source, wiring, connectors, fusing, and installation for their deployment.

# Root cause and symptoms

Tracker One includes a USB-C connector that can be used for flashing, debugging, and 5V power.

The issue addressed by this advisory is not a defective Tracker One USB connector. The issue is that the device cannot determine the current-carrying capability, condition, temperature rise, wear, length, or resistance of every customer-supplied USB-A to USB-C cable, adapter, extension, charger, or field installation. As such, the device might pull more power from this cable than the cable manufacturer designed for. This is not related to any defect of the Tracker One - USB cables do not carry this information electronically and thus, Tracker One can not determine the electronic properties or condition of the attached USB cable.

Particle's power documentation for the Tracker One describes its capabilities as including USB DPDM/BC1.2 source detection and the Tracker One's power-source current limits. These mechanisms determine available source capability; they do not qualify the physical cable and connector path selected by the customer. The sourced cable must be capable of the 1.5A source current.

## Background

Higher-resistance USB cables or connectors can dissipate heat when current flows through them. As current is increased, heating increases approximately as:

Power dissipated as heat =  $I^2R$

For example, reducing USB/input current from 1500 mA to 900 mA reduces resistive heating in the same cable/connector path to approximately:

$$(900 / 1500)^2 = 36\%$$

This is approximately a 64% reduction in resistive heating for the same resistance.

Cables that were not designed for 1.5A that are used to carry it can therefore dissipate the current as heat, reducing the efficiency of the available changing current significantly.

Observed symptoms may include:

- USB connector or cable becoming unusually warm or hot
- Discoloration or deformation of the USB connector, cable, or enclosure area
- Intermittent charging
- Charge fault indication
- Device resets or brownouts under load
- Slower than expected charging
- VBUS/input voltage sag under load

## Recommended customer action

All Tracker One customers should review their USB power and charging configuration.

Customers should determine:

1. Whether Tracker One devices are powered or charged by USB in development, production, service, or field operation.
2. Whether the USB cable is Particle-supplied, customer-supplied, field-replaced, extended, or otherwise modified.
3. Whether the USB power source is a computer USB port, USB charger, vehicle adapter, hub, battery pack, or other 5 V supply.
4. Whether the cable and source have been qualified for the configured Tracker One input current.
5. Whether the installation has been validated under worst-case operating conditions, including ambient temperature, enclosure mounting, cable routing, vibration, and battery state.

Customers should not assume that a USB power source capable of high current means that the USB cable and connector path are also suitable for sustained high-current charging.

If the USB cable or field power configuration is unknown, cannot be controlled, or cannot be qualified, Particle recommends reducing the Tracker One charging/input current or using a known-good external power input appropriate for the deployment.

# Issue mitigation

## USB CABLE AND POWER SOURCE QUALIFICATION

Customers should use only USB cables, adapters, and power sources that are appropriate for the configured current and installation environment.

Recommended practices:

- Use a known-good USB cable with an appropriate current rating of 1.5A.
- Avoid damaged, worn, bent, loose, corroded, modified, or unmarked USB cables.
- Avoid long USB cable runs unless the cable has been qualified for voltage drop and heating.
- Avoid unqualified USB vehicle adapters or third-party charging accessories.
- Validate connector temperature rise under worst-case conditions.
- Inspect field installations where cables may be replaced by operators, installers, or end users.
- Do not use a USB cable or connector that shows signs of excessive heating.

The Particle-supplied USB-A to USB-C cable is intended for development and setup. Customers must determine whether that cable, or any alternate cable used in production, service, or field operation, is suitable for the customer's operating environment and configured charge/input current.

## CURRENT CONFIGURATION

It is possible to reduce the current input on Tracker One to utilize current sources down to 500mA if the external USB cable is unknown in its rating.

By default, Tracker Edge currently configures Tracker One with the following configuration:

```
commonCfg.chargeCurrentHigh = 1024; // milliamps
commonCfg.inputCurrent = 1500;      // milliamps
```

These values are present in the Tracker Edge `TrackerOneConfiguration.h` default configuration.

Customers should configure charging/input current based on the actual USB cable and power source used in their deployment.

Recommended profiles based on the USB cable source:

| Use case                | Battery charge current | USB/input current | Notes                                                                                                                                        |
|-------------------------|------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum charge current  | 1024 mA                | 1500 mA           | Only where the cable, connector path, and power source have been qualified for sustained operation.                                          |
| Reduced-current profile | 512 mA                 | 900 mA            | Recommended where the cable/source is not fully controlled but is expected to support at least 900 mA. Reduces connector heating risk.       |
| 500 mA USB path only    | 512 mA                 | 500 mA            | Use only if the source/cable path is limited to 500 mA. Charging may be very slow or may pause depending on system load. Validate carefully. |

Do not configure the Tracker One input current higher than the qualified capability of the cable, connector path, and power source.

Particle's power documentation lists valid `powerSourceMaxCurrent()` values of 100, 150, 500, 900,

1200, and 1500 mA, and describes that actual charge current is limited by both the charge current setting and input current limit.

#### RECOMMENDED REDUCED-CURRENT FIRMWARE SETTING

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For customers who cannot confirm that all deployed USB-A to USB-C cables and power sources are suitable for the higher-current default, Particle recommends using the **500 mA USB path only** setting.

The 512 mA charge-current setting minimizes the battery charge current. The 900 mA input-current setting provides additional current margin above the battery charge-current setting for system operation while charging.

Customers with a known 500 mA USB source or cable path should not use the 900 mA input-current setting unless the cable and source have been qualified for 900 mA operation.

#### CONFIGURATION USING PARTICLE CONSOLE ENVIRONMENT VARIABLES

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For devices running a Device OS version that supports power configuration through environment variables, the reduced-current profile can be configured without a user-firmware update.

Set the following environment variables for the affected product or devices:

| Environment variable         | Reduced-current value |
|------------------------------|-----------------------|
| PARTICLE_PMIC_CHARGE_CURRENT | 512                   |
| PARTICLE_PMIC_INPUT_CURRENT  | 900                   |

For known 500 mA USB-only installations, use:

| Environment variable         | 500 mA USB-path value |
|------------------------------|-----------------------|
| PARTICLE_PMIC_CHARGE_CURRENT | 512                   |
| PARTICLE_PMIC_INPUT_CURRENT  | 500                   |

After changing environment variables, roll out the configuration to the target devices and validate behavior on a controlled subset of the fleet before broad deployment. Particle's environment-variable documentation notes that variables can be edited and rolled out to devices, and that Device OS environment-variable support requires Device OS 6.4.0 or later; customers should confirm the Device OS version required for the specific power environment variables before using this mitigation.

Note that environment variables are only supported in Tracker Edge Firmware v23 and later, which require Device OS 6.5.0 or later.

#### ALTERNATIVE POWER INPUT

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For production deployments where USB cable quality cannot be controlled, customers should consider powering Tracker One through the M8/VIN input using a properly designed and qualified power installation.

Tracker One documentation describes a VIN input option through the M8 connector, including 6 to 30 VDC at 2 A on the M8 connector.

## Charging time impact

Reducing charge current will increase battery charging time.

Approximate bulk charging time can be estimated as:

Charging time in hours  $\approx$  battery capacity to replace in mAh / charge current in mA

For a 3000 mAh battery, ideal 0–100% bulk charging time is approximately:

| Charge current | Ideal bulk charging time |
|----------------|--------------------------|
| 1024 mA        | 2.9 hours                |
| 512 mA         | 5.9 hours                |
| 500 mA         | 6.0 hours                |

Actual full charge time will be longer because lithium battery charging slows near full charge, the Tracker One consumes power while operating, charging may be temperature-limited, and the input current limit may reduce available charge current.

For Tracker One devices that are normally powered continuously, this increased charging time may not affect normal operation. For devices that only receive power for short windows, customers must validate that the reduced charge current still meets their application requirements.

The alternative to impacting charge time is to utilize a known 1.5A capable USB cable.

# Who should make a change?

Customers should reduce the charge/input current if any of the following are true:

- The USB cable type used in the field is unknown.
- The USB cable may be replaced by installers, operators, service technicians, or end users.
- The cable is customer-supplied or sourced from multiple suppliers.
- The cable is long, worn, damaged, or exposed to vibration or harsh environments.
- The installation uses USB vehicle adapters or third-party high-current USB chargers.
- The customer cannot confirm that the full USB cable and connector path is suitable for sustained 1.5 A operation.
- Any USB connector heating has been observed.

Customers may retain higher-current settings only if the USB cable, connector path, and power source have been qualified for the configured current and the installation has been validated under expected operating conditions.

## Revision history

| Date       | Notes         |
|------------|---------------|
| 2026-06-08 | Initial draft |
| 2026-09-18 | Release       |